

Thesis Plan

Decentralized Multi-Agent Reinforcement Learning for Power Grid Topology Control

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Chapter 1

Thesis Plan

1.1 Plan Inspiration

This plan follows the structure of the reference thesis in `mAI2024deMolB.pdf`: start from the power-grid motivation and research questions, introduce the scientific background, describe the environment before the algorithms, then separate methods from results and close by answering the research questions. The adapted structure below is tuned to this repository: Grid2Op topology control, decentralized MAPPO, graph encoders, rerun ablations, and sparse intervention mechanisms.

1.2 Working Title

Decentralized Multi-Agent Reinforcement Learning for Sparse Power Grid Topology Control

1.3 Central Thesis Question

How can decentralized multi-agent reinforcement learning agents learn reliable power-grid topology control policies that preserve survival performance while scaling to combinatorial action spaces and avoiding unnecessary interventions?

1.4 Research Questions

1. How should the Grid2Op topology-control problem be formulated as a decentralized multi-agent decision problem?
2. Which architectural choices improve survival and generalization: flat MLP policies, graph neural network encoders, shared actor weights, critic design, or optimized critic updates?
3. How do initialization and training choices, especially the initial do-nothing bias and reward normalization, affect exploration and learned behavior?

4. Can sparse-control mechanisms, such as rho-based overrides, intervention gates, fixed penalties, or adaptive intervention budgets, reduce unnecessary topology actions without sacrificing episodic survival?

1.5 Expected Contributions

- A clear decentralized formulation of topology optimization in Grid2Op.
- A controlled comparison of MAPPO baselines, GNN encoders, critic variants, and initialization choices.
- An ablation study of sparse intervention mechanisms for more operator-like control.
- A reproducible experiment and analysis pipeline based on W&B histories, cached metrics, notebooks, and final evaluation scripts.

1.6 Chapter Outline

Chapter 1: Introduction

Goal. Motivate topology control as a low-cost but difficult operational tool for power-grid reliability.

Planned content.

- Energy-transition context: higher demand, renewable uncertainty, and stressed transmission networks.
- Topology control as an alternative or complement to redispatching and grid expansion.
- Why the problem is hard: sequential control, physical constraints, rare failures, and combinatorial topology actions.
- Why decentralized MARL is a natural fit: local substations, regional decision making, reduced per-agent action spaces, and scalability.
- Research questions, contributions, and thesis outline.

Chapter 2: Background and Related Work

Goal. Give the reader enough power-systems and RL background to understand the design choices.

Planned content.

- Power-grid operation: substations, busbars, line loading, overloads, contingencies, and remedial actions.
- Grid2Op and the Learning to Run a Power Network setting.
- Reinforcement learning basics: MDPs, policy gradients, PPO, and MAPPO.
- Multi-agent reinforcement learning: decentralized execution, centralized training, credit assignment, and non-stationarity.

- Graph neural networks for grid-structured observations.
- Safe, sparse, and constrained RL ideas relevant to intervention budgets.

Chapter 3: Environment and Problem Formulation

Goal. Define exactly what the agents observe, choose, optimize, and are evaluated on.

Planned content.

- Grid2Op environment used in the repository and the selected chronics.
- Multi-agent decomposition: agent-region or agent-substation assignment.
- Observation spaces: local grid state, line loadings, topology state, and optional graph observations.
- Action spaces: local topology actions, action 0 as do nothing, legal action constraints, and action-space reduction.
- Reward and evaluation metrics: episodic survival, illegal action rate, action-0 fraction, non-idle fraction, and train/test chronic split.
- Reproducibility setup: seeds, W&B logging, cached histories, and deterministic evaluation.

Chapter 4: Methods

Goal. Present the implemented learning systems and the ablations.

Planned content.

- MAPPO baseline: actor, critic, rollout collection, PPO objective, and centralized training assumptions.
- Baseline MLP agents and the A0 family of reruns.
- Graph-encoder agents: bus graph construction, GINE/GAT/GCN variants, node and edge pre-encoders, graph readout, and flat-feature concatenation.
- Critic variants: GNN critic versus MLP critic, legacy versus optimized critic updates.
- Initialization and exploration choices: entropy coefficient, entropy decay, and initial do-nothing probability.
- Sparse-control mechanisms: evaluation-time rho heuristic, intervention gate, fixed sparse-action penalties, and adaptive intervention budgets.
- Experimental protocol for ablations: controlled seeds, fixed evaluation cadence, and fair baseline comparisons.

Chapter 5: Experiments and Results

Goal. Show which design choices matter and what behaviors the agents learn.

Planned content.

- Main baseline performance: training stability and test survival.

- GNN architecture ablations: concat-flat, actor sharing, GINE versus GAT or weighted GCN, light versus full encoders, and critic design.
- A0 rerun ablations: optimized critic updates, smaller MLP, reward normalization, and do-nothing initialization bias.
- Sparse-control ablations: fixed penalty, global-safe weighting, local-safe adaptive budget, rho-threshold sensitivity, and budget target sensitivity.
- Behavioral analysis: how often agents act, whether interventions are local, and whether high survival comes with excessive topology churn.
- Generalization: seen versus unseen chronics and robustness across seeds.

Figures and tables to prepare.

- Table of run families and what each ablation changes.
- Survival curves for baseline, GNN, A0, and sparse-control comparisons.
- Action-frequency plots: action 0 fraction, any non-idle action, and multi-agent simultaneous interventions.
- Summary table with final survival, peak survival, and action sparsity.

Chapter 6: Discussion

Goal. Interpret the results rather than just restating them.

Planned content.

- Answer each research question directly.
- Explain which components appear essential, useful, or unnecessary.
- Discuss the tradeoff between survival and sparse interventions.
- Relate learned behavior to operator-like control: act rarely, act locally, and act when the grid is unsafe.
- Discuss limits: simulator scope, incomplete operational constraints, seed variance, training cost, and dependency on chosen chronics.

Chapter 7: Conclusion and Future Work

Goal. Close with the main thesis claim and next research directions.

Planned content.

- Short summary of the problem, approach, and core findings.
- Final conclusion about decentralized MARL for topology control.
- Future work: richer graph representations, larger grids, outage contingencies, constrained RL tuning, better coordination mechanisms, and operational validation.

1.7 Appendix Plan

- Appendix A: Environment details and action-space construction.
- Appendix B: Hyperparameters and training configuration tables.
- Appendix C: Full ablation run registry.
- Appendix D: Additional survival and behavior plots.
- Appendix E: Reproducibility notes for notebooks, cached W&B histories, and evaluation scripts.

1.8 Immediate Writing Order

1. Write Chapter 3 first, because it fixes notation and the experiment vocabulary.
2. Write Chapter 4 next, using the existing configuration READMEs as source notes for method variants.
3. Draft the Chapter 5 table of ablations before prose, so the results chapter has a stable spine.
4. Write Chapter 1 once the main conclusion is clearer.
5. Finish Chapters 6 and 7 after the final plots and summary metrics are frozen.

1.9 Open Decisions

- Confirm the final thesis title and whether to emphasize GNNs, sparse-control, or decentralized MARL most strongly.
- Decide which experiment families are final and which are only exploratory.
- Decide whether the main claim is performance improvement, improved sparsity at equal survival, or reproducible ablation insight.
- Choose the final bibliography style required by the university.

Chapter 2

Preliminary Results

2.1 Rerun Episodic Survival Ablations

This section summarizes the rerun survival plots generated from the episodic survival rerun notebook. The metric is test episodic survival in percent. Thick lines show the seed-aggregated mean curve, faint lines show individual seed curves, and shaded regions show the variability band used in the notebook plots.

2.1.1 Result Summary

The reruns suggest three main trends. First, the strongest GINE rerun is the light A0-style GINE with an MLP critic and optimized critic updates (**best_14**), which reaches a final survival of about 95.7%. Second, the A0 MLP baseline is robust: **rerun_a0_opt**, **rerun_a0_nono**pt, and the smaller **rerun_a01** variant all finish around 93% survival, while disabling reward normalization (**rerun_a02**) ends lower. Third, the initial do-nothing bias has a large but non-monotonic effect in these reruns: **rerun_bias_05** performs poorly, while **rerun_bias_07** reaches nearly perfect final survival.

Table 2.1: Selected final and peak survival values extracted from the rerun survival notebook figures.

Experiment group	Run	Final survival (%)	Peak survival (%)
GINE rerun	best_00 baseline	59.2	93.9
GINE rerun	best_10 light GINE MLP critic	78.9	90.4
GINE rerun	best_12 optimized critic, bias 0.0	82.5	85.7
GINE rerun	best_14 light A0-style GINE MLP critic	95.7	97.4
A0 core	rerun_a0_opt	93.4	98.6
A0 core	rerun_a0_nono pt	93.1	95.8
A0 core	rerun_a01 smaller MLP	93.7	100.0
A0 core	rerun_a02 no reward normalization	81.8	97.1
A0 core	old_a0_nono pt_ noval20	22.3	68.4
A0 bias	rerun_bias_05	50.8	70.2
A0 bias	rerun_bias_07	99.3	100.0

2.1.2 GINE best_00 Versus best_10–best_14

Figure 2.1 compares the historical **best_00** GINE baseline against five controlled reruns. Removing the initial do-nothing bias alone (**best_11**) does not improve the final curve. The optimized-critic variants are more competitive, and the strongest curve is **best_14**, which combines a lighter GINE actor, MLP critic, A0-style entropy and initialization choices, no flat concatenation, and optimized critic updates.

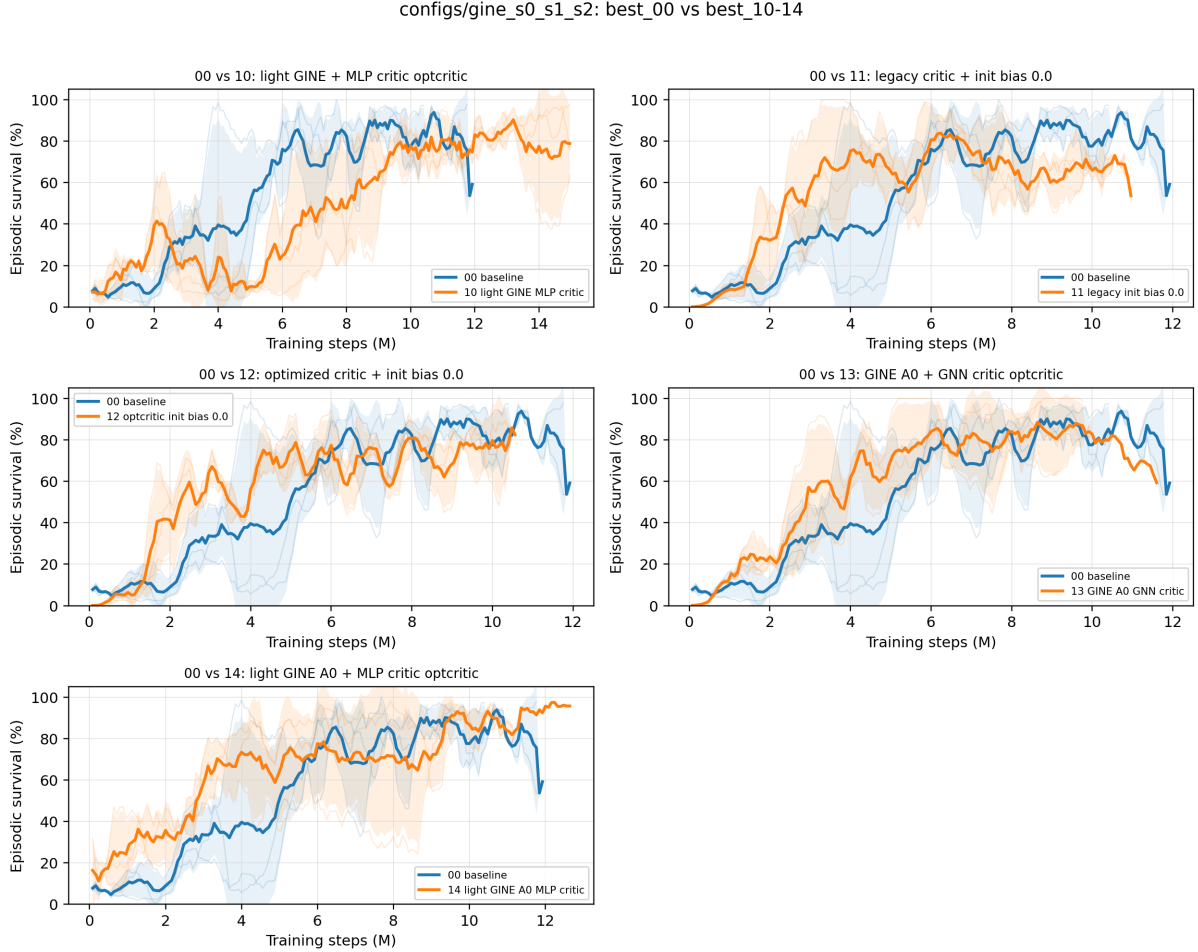


Figure 2.1: Seed-aggregated test episodic survival for GINE **best_00** against **best_10**–**best_14**.

2.1.3 A0 Core Reruns

Figures 2.2 and 2.3 show the A0 core reruns. The optimized baseline, non-optimized critic update, and smaller MLP variant all reach similar final survival. The no-reward-normalization run learns more slowly and finishes lower, while the older non-optimized no-validation baseline is clearly not competitive under this comparison.

2.1.4 A0 Initial Do-Nothing Bias

Figure 2.4 isolates the initial do-nothing bias. The **rerun_bias_05** curve remains unstable and substantially below the zero-bias baseline, whereas **rerun_bias_07** eventually reaches the

A0 core reruns: rerun_a0_opt baseline pairwise comparisons

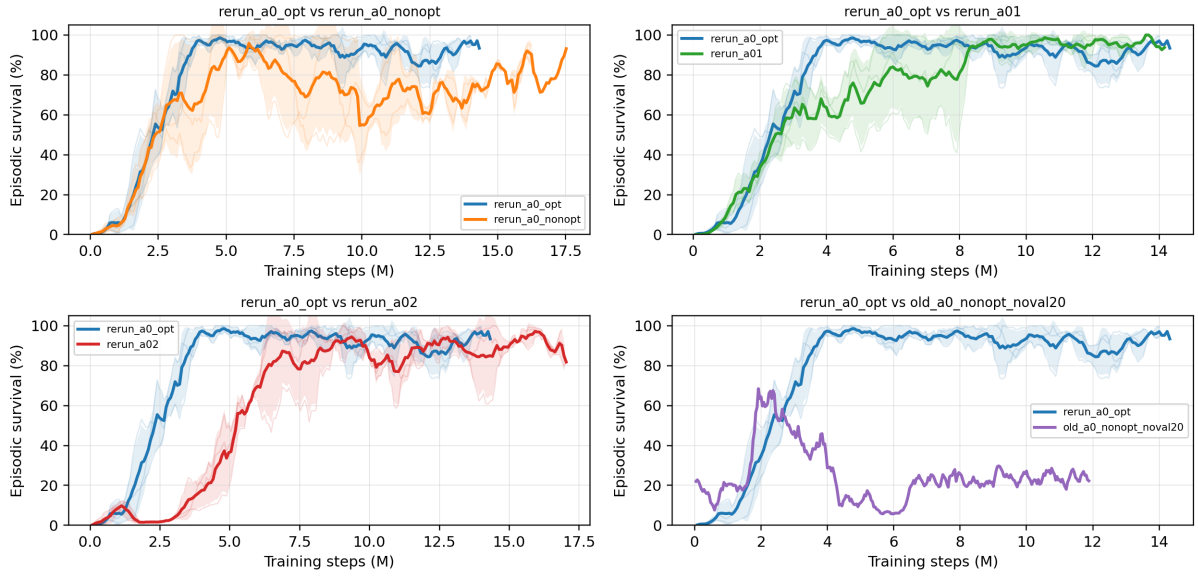


Figure 2.2: Pairwise A0 rerun comparisons against the `rerun_a0_opt` baseline.

A0 core reruns: `rerun_a0_opt` vs `rerun_a0_nonopt` / `rerun_a01` / `rerun_a02` / old noval20 nonopt

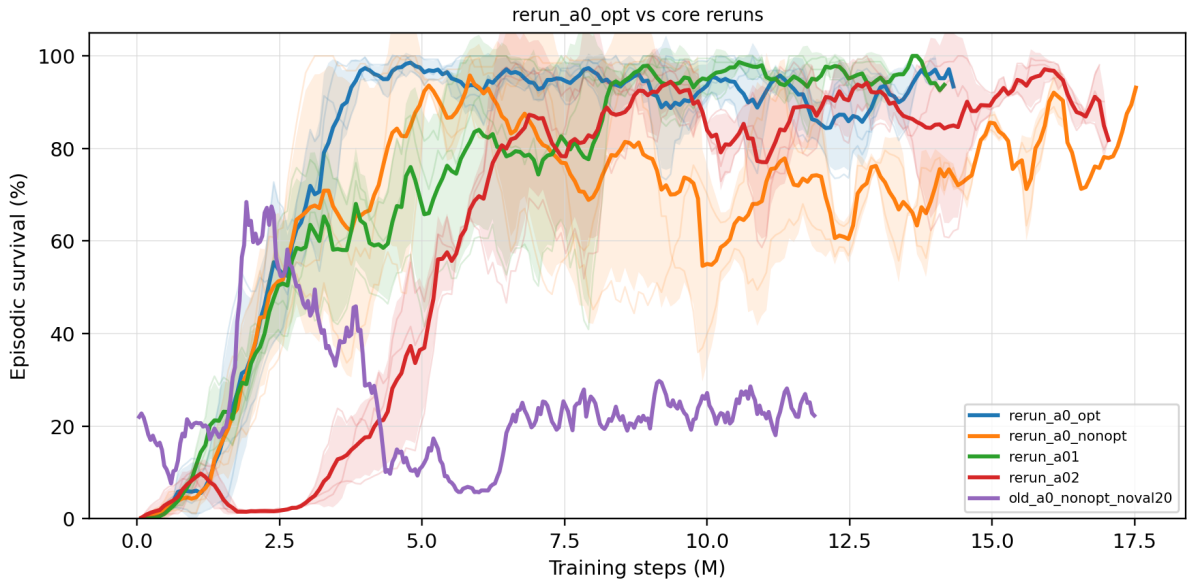


Figure 2.3: All A0 core rerun survival curves shown on one axis.

highest final survival in this comparison. The conclusion is therefore not that larger bias is always better, but that the initialization prior strongly shapes exploration and should be treated as a first-order ablation variable.

rerun_a0_opt baseline vs rerun_bias_05 / rerun_bias_07

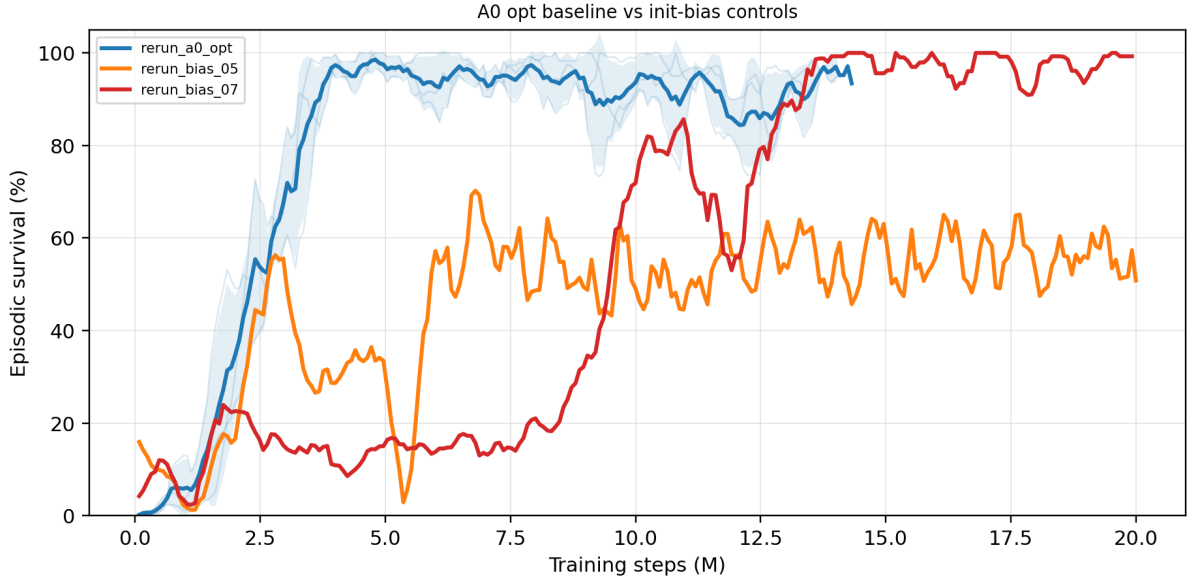


Figure 2.4: A0 baseline compared with initial do-nothing bias controls `rerun_bias_05` and `rerun_bias_07`.

2.1.5 Ablation Takeaway

Overall, these reruns indicate that the strongest current direction is not a single isolated modification, but the combination of stable A0-style training choices with architectures that keep the critic and graph encoder simple enough to optimize reliably. Reward normalization and initialization bias both have large effects on survival, while optimized critic updates and lighter GINE/MLP critic variants can improve the final GINE rerun performance.

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